Attack-Method Choice Changes Robustness Conclusions for a Standard LSTM Rainfall-Runoff Model

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Key Points:

- Iterative adversarial perturbations (Auto-PGD) reveal 17.5× greater median vulnerability than Gaussian random noise for a 531-basin CAMELS-US benchmark LSTM at $\varepsilon = 0.1$.
- NSE-based headline statistics use 514 finite basin metrics; one-third of these basins fall below NSE = 0 under iterative attack at observation-error-scale perturbation budgets.
- Statistically undetectable perturbations that preserve input-feature means and standard deviations still cause substantial degradation, indicating that routine quality control is insufficient.

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Abstract

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong predictive skill in rainfall-runoff modeling. As these models move toward operational deployment, their robustness to input uncertainty becomes a critical concern. Yet existing assessments often rely on random noise or single-step adversarial methods (FGSM), which may underestimate worst-case vulnerability. Here we present a multi-level adversarial stress test of a 531-basin CAMELS-US temporal benchmark LSTM, employing a spectrum of perturbation methods from random noise through iterative optimization (Auto-PGD), three constraint levels of increasing plausibility, and targeted attacks on flood and pre-event windows. These perturbations are not intended to reproduce the empirical distribution of real-world forcing errors; rather, they serve as constrained worst-case stress tests within observation-error-scale uncertainty budgets. We find that this model is substantially more fragile than random-noise testing suggests: at $\varepsilon = 0.1$, Auto-PGD causes a median NSE degradation of 0.41 across 514 finite basin metrics, compared to 0.02 for Gaussian noise, an adversarial-to-random ratio of 17.5:1. FGSM gives a similar but weaker median degradation of 0.36, indicating that the largest conclusion change is between random and gradient-based testing, while iterative optimization still strengthens the attack. Under Auto-PGD, 167 of 514 finite basins (32.5%) fall below NSE = 0. Furthermore, perturbations that preserve the statistical properties of input features (mean and standard deviation) still cause significant degradation (median ΔNSE $= -0.19$), suggesting that standard quality-control procedures cannot detect all harmful input-error patterns. These results show that robustness assessment for standard hydrological deep-learning configurations requires multi-level stress testing: random-noise sensitivity analysis substantially underestimates the vulnerability that gradient-based adversarial attacks reveal.

Plain Language Summary

LSTM neural networks are increasingly used for streamflow prediction and are moving toward operational use. We tested how sensitive a standard 531-basin CAMELS-US benchmark LSTM rainfall-runoff model is to small but structured errors in its weather inputs within realistic uncertainty bounds. Using an optimization method called Auto-PGD, we searched for the most harmful input errors allowed by those bounds. These perturbations are not meant to mimic the most common real-world errors; they are stress tests that ask how badly the model can fail if the input errors happen to align with its sensitivities. Across 514 basins with finite NSE metrics, these constrained worst-case perturbations were about 17.5 times more damaging than Gaussian random noise of the same magnitude. The commonly used FGSM attack found much of the same vulnerability at $\varepsilon = 0.1$, but remained weaker than iterative Auto-PGD, especially at larger perturbation budgets. Nearly one in three finite basins fell below NSE = 0 under iterative attack. We also found that some harmful perturbations would pass standard data-quality checks because they preserve the mean and standard deviation of the input variables. These results do not imply that all LSTM models should not be deployed, but they show that standard hydrological deep-learning configurations need stronger stress testing than random-noise sensitivity analysis alone.

1 Introduction

Long Short-Term Memory (LSTM) networks have transformed hydrological prediction over the past decade. Beginning with the pioneering work of Kratzert et al. (2018), deep learning models have consistently matched or exceeded the performance of process-based hydrological models on standardized benchmarks such as CAMELS (Addor et al., 2017), achieving median Nash-Sutcliffe Efficiency (NSE) values above 0.7 across hundreds of catchments (Kratzert et al., 2019; Gauch et al., 2021). These advances have catalyzed growing interest in deploying LSTM-based models in operational forecasting systems (Nearing et al., 2021).

As deployment moves forward, a fundamental question demands attention: how reliable are these models when confronted with imperfect input data? In practice, meteorological forcings are never exactly known. Rain gauge measurements carry systematic biases of 5–15% (Sevruk, 1982; Groisman & Legates, 1994). Gridded products such as Daymet, ERA5, and CHIRPS introduce additional uncertainty through spatial interpolation, with temperature errors of 1–2 °C and precipitation errors that can exceed 2 mm/day in complex terrain (Thornton et al., 2021; Hersbach et al., 2020). If model predictions are highly sensitive to such perturbations, then predictive accuracy on clean benchmarks may overstate operational reliability.

The conventional approach to testing input sensitivity, adding random Gaussian noise to the forcings, provides a measure of average-case robustness. However, random perturbations distribute their probe equally in all directions and are unlikely to land on the particular forcing-error pattern to which the trained model is most sensitive. Adversarial robustness analysis, borrowed from the machine-learning security literature (Goodfellow et al., 2015; Madry et al., 2018), offers a more systematic alternative: it uses gradient information from the trained model itself to identify, within a given uncertainty budget, the forcing perturbation that would most damage the streamflow prediction, and so provides a lower bound on worst-case performance. In this study, such perturbations are not treated as literal samples from the empirical forcing-error distribution. They are constrained stress tests that ask whether harmful error patterns exist at all within plausible uncertainty budgets, not whether such patterns dominate observed forcing errors.

Yang et al. (2026) applied adversarial robustness analysis to hydrological models using the Fast Gradient Sign Method (FGSM; Goodfellow et al., 2015), evaluating LSTM and HBV models across 1,347 German catchments from the CAMELS-DE dataset. Their findings were encouraging: LSTMs demonstrated greater robustness than HBV models, catastrophic failure was rare, and model responses scaled approximately linearly with perturbation magnitude.

However, FGSM is a single-step attack: it evaluates the model's gradient once at the clean forcing and takes one maximum-sized push along the resulting sign direction. While computationally efficient, single-step methods are known in the adversarial machine-learning literature to systematically overestimate model robustness, because they implicitly linearize the model's response to forcing perturbations around the clean operating point. Iterative methods such as Projected Gradient Descent (PGD; Madry et al., 2018) and Auto-PGD (Croce & Hein, 2020) refine the perturbation over many smaller steps and so capture the nonlinearities in the model's response, consistently finding more harmful forcing patterns than single-step probes. The gap between single-step and iterative methods is not merely quantitative; it can qualitatively change conclusions about whether a model is "robust" or "vulnerable" (Carlini et al., 2019). Yang et al. (2026) themselves acknowledged this limitation, noting the need for "additional test scenarios and perturbation methods."

This raises a fundamental question: **how fragile is a standard LSTM rainfall-runoff configuration when subjected to systematic, multi-level adversarial stress testing?**

To answer this question, we construct a multi-level adversarial stress-testing framework and apply it to a standard LSTM model on the 531-basin CAMELS-US temporal benchmark. The framework comprises three components: (i) a *perturbation spectrum* spanning five methods of increasing optimization strength, from random noise through single-step FGSM to iterative Auto-PGD; (ii) a *constraint hierarchy* controlling perturbation plausibility, including a statistical constraint that preserves input-feature means and standard deviations; and (iii) *operationally motivated analyses*, a Carlini-Wagner (C&W) attack quantifying the minimum perturbation to break the model, targeted flood/low-flow attacks, and a causal trigger analysis restricting perturbations to pre-event windows.

While Yang et al. (2026) focused on comparing model architectures under a single attack method, our study takes a complementary perspective: we hold the model fixed and systematically vary the attack to characterize how the strength and structure of the stress test affect the conclusions drawn about model robustness.

2 Methods

2.1 Study Area and Target Model

We evaluate a single-layer LSTM model (Kratzert et al., 2018) with 128 hidden units, implemented in the neuralhydrology framework (Kratzert et al., 2022). We deliberately use this standard configuration as the victim model so that the analysis isolates attack-method effects rather than architectural differences. The model was trained on the CAMELS-US dataset (Newman et al., 2015; Addor et al., 2017) with five dynamic input features from the Daymet forcing product: precipitation (mm/day), shortwave radiation (W/m^2), maximum temperature (°C), minimum temperature (°C), and vapor pressure (Pa). It also uses 14 static catchment attributes.

The target model is a local reproduction of the Kratzert et al. 531-basin temporal benchmark: the same 531 basin list is used for training, validation, and testing, with disjoint time periods. The training period is 1 October 1990–30 September 1995, validation is 1 October 1995–30 September 2000, and testing is 1 October 2000–30 September 2005. Adversarial evaluation pins the strongest saved checkpoint (epoch 20) from a 30-epoch training run; reproducibility details and run paths are provided in Supporting Information Section S2.

2.2 Perturbation Methods

We employ five perturbation methods that span a spectrum of increasing optimization strength (Table 1). All methods operate on the standardized forcing time series, with each variable rescaled to zero mean and unit variance over the training period. The perturbation budget ε caps the shift, in standardized units, of any single timestep–variable component. Table 2 translates ε back into physical units (mm/day, °C, etc.).

This design is diagnostic rather than predictive. The random and adversarial perturbations are not intended to reproduce the empirical distribution of operational forcing errors; they test whether, within a specified uncertainty budget and plausibility constraints, structured input errors that materially degrade model performance can be constructed at all.

2.2.1 Random Noise Baselines

Three random perturbation methods serve as baselines, each representing an idealized form of forcing error. *Gaussian noise* draws independent samples from $\mathcal{N}(0, \varepsilon^2)$ clipped to $[-\varepsilon, \varepsilon]$ and represents white sensor noise that varies independently from one timestep and variable to the next. *Multiplicative bias* scales each feature by a random factor drawn from $\mathcal{U}(1-\varepsilon, 1+\varepsilon)$ and represents persistent calibration error, such as a miscalibrated rain gauge or radiation sensor whose reading is uniformly off by a small fraction throughout the period. *Temporally correlated noise* draws from an AR(1) process with lag-1 autocorrelation of 0.7, rescaled to respect the ε bound, and represents errors that propagate across consecutive days—for example, a gridded forcing product that systematically misrepresents the same weather system over its lifetime. None of the

Table 1 — Perturbation methods used in this study, ordered by optimization strength.

Method	Type	Gradient	Iter.	Role
Gaussian noise	Random baseline	None	0	Traditional noise sensitivity
Mult. bias	Random baseline	None	0	Systematic sensor bias
Temp. corr. noise	Random baseline	None	0	Realistic temporal error
FGSM	Single-step adv.	Single	1	Bridge; used by Yang et al. (2026)
Auto-PGD	Iterative adv.	Iterative	50	Worst-case lower bound
C&W regression	Min. perturbation	Iterative	200	Safety margin quantification
Causal trigger	Temporally constr.	Iterative	100	Pre-event vulnerability

three uses gradient information from the model; together they characterize the average-case sensitivity of the trained model to perturbations of the corresponding statistical structure.

2.2.2 FGSM (Single-Step Adversarial)

The Fast Gradient Sign Method (FGSM) seeks the worst single-step perturbation within the budget. Backpropagation through the trained model returns, at every timestep and for every forcing variable, the partial derivative of the loss with respect to that input value. The sign of this derivative indicates whether shifting the value upward or downward degrades the prediction; FGSM applies a perturbation of magnitude ε in this unfavorable direction at every component simultaneously,

$$\mathbf{x}_{\mathrm{adv}} = \mathbf{x} + \varepsilon \cdot \mathrm{sign}(\nabla_{\mathbf{x}}\mathcal{L}(f(\mathbf{x}), \mathbf{y}))\,, \tag{1}$$

where $\mathcal{L}$ is the negative NSE loss, f is the trained model, and $\mathbf{y}$ is the observed streamflow. FGSM requires a single forward pass to evaluate the loss and a single backward pass to obtain the gradient, and produces a deterministic perturbation per input sequence. Because the magnitude is uniform and the direction is set by the gradient at the clean input, FGSM corresponds to the worst forcing-error pattern that a linearization of the model around the clean input would predict.

2.2.3 Auto-PGD (Iterative Adversarial)

FGSM implicitly linearizes the model around the clean forcing. For small budgets this linearization is accurate, and a single sign step nearly saturates the achievable damage. For larger budgets the model's response curves away from the linear approximation within the ε neighborhood, so the direction picked at the clean input is no longer the worst direction within the budget. Auto-PGD (Croce & Hein, 2020) addresses this by iterating: rather than one step of size ε, it takes many smaller steps, recomputes the gradient at each new candidate forcing, and projects the result back into the budget whenever a component would otherwise leave it,

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{B}(\mathbf{x},\varepsilon)}\left[\mathbf{x}_t + \alpha_t \cdot \mathrm{sign}(\nabla_{\mathbf{x}}\mathcal{L}(f(\mathbf{x}_t), \mathbf{y}))\right]. \tag{2}$$

Here $\Pi_{\mathcal{B}(\mathbf{x},\varepsilon)}$ denotes projection onto the ε-neighborhood around the clean input, implemented by clipping any timestep–variable component that exceeds the budget back to the boundary, and α_t is an adaptive step size that starts at a fraction of ε and decreases as the attack converges. We use 50 iterations and one restart from a random initialization within the ε-neighborhood, which mitigates poor local optima at modest additional cost. Whereas FGSM corresponds to the worst pattern predicted by a linearization of the model, Auto-PGD identifies the worst forcing-error pattern attainable under the actual nonlinear model within the same budget.

2.2.4 C&W Regression Attack

Whereas FGSM and Auto-PGD quantify the maximum damage achievable under a fixed perturbation budget, the Carlini–Wagner (C&W) attack (Carlini & Wagner, 2017), adapted here for regression, quantifies the dual quantity: the minimum perturbation magnitude required to drive model performance below a specified threshold. This converts the analysis from an attack-magnitude axis to a per-catchment safety-margin axis. Catchments that require a large L_2 perturbation before NSE drops below zero have a wide margin against forcing error of any structure; catchments broken by a small L_2 perturbation are operationally fragile.

We minimize $\|\boldsymbol{\delta}\|_2$ subject to $\mathrm{NSE}\big(f(\mathbf{x} + \boldsymbol{\delta}), \mathbf{y}\big) \; < \; 0$ using a Lagrangian relaxation, where a single weight trades perturbation size against constraint satisfaction. The inner minimization runs 200 iterations of gradient descent at learning rate 0.01; the weight is selected by a 5-step binary search. The output for each catchment is the smallest L_2 forcing perturbation, in standardized units, that drives the test-period NSE below zero.

2.2.5 Causal Trigger Attack

The causal trigger attack quantifies how much of the unrestricted iterative damage is achievable by perturbations confined to the period most relevant for flood early warning. The days immediately preceding a flood peak are operationally consequential: forecast skill is most valuable in this window, and input data are most often degraded by sensor failure or telemetry latency. We identify flood peaks as local maxima of observed discharge that exceed the 90th percentile of the test-period flow and restrict perturbations to a window of 1, 3, 7, or 14 days before each identified peak. Within the window, perturbations are optimized iteratively for 100 steps using the same gradient-projection scheme as Auto-PGD; outside the window, the forcing remains unperturbed. Comparing the four window lengths quantifies the share of vulnerability concentrated in the immediate pre-event period.

2.3 Constraint Hierarchy

The same ε budget can describe forcing-error patterns of very different physical realism. To make this explicit, we evaluate Auto-PGD under three nested constraints, ordered from the most permissive to the most restrictive.

The first level, the *magnitude constraint*, only requires that no single timestep–variable component be perturbed by more than ε in standardized units. This is the default constraint used throughout the perturbation literature; it admits forcing patterns that may be physically implausible (negative precipitation, unrealistically extreme temperatures) so long as each individual deviation stays within the budget.

The second level, the *physical constraint*, additionally clips precipitation to non-negative values and constrains temperatures to climatologically reasonable ranges, following Yang et al. (2026). This rules out patently unphysical forcing patterns while leaving the magnitude budget unchanged.

The third level, the *statistical constraint*, additionally requires the perturbed forcing series to preserve, for each variable, the same mean and standard deviation as the clean series over the evaluation period. This is intended as a proxy for input data that has already passed routine quality-control checks comparing summary statistics to climatology: any QC pipeline based on per-variable means and standard deviations would, by construction, fail to flag perturbations satisfying this constraint.

2.4 Mapping ε to Physical Units

Table 2 maps ε values to physical units using training-period standard deviations. At $\varepsilon = 0.1$, perturbations correspond to approximately ± 0.8 mm/day for precipitation and ± 1.1 °C for temperature, values within the typical error range of gridded products such as Daymet (Thornton et al., 2021). Our primary analyses focus on $\varepsilon = 0.1$ as a realistic worst-case scenario.

Table 2 — Mapping between ε and physical units for each input feature. Training-period standard deviations (σ) are used for conversion.

Feature	σ	ε=0.05	ε=0.1	ε=0.2	Typical obs. error
Precip. (mm/day)	7.73	±0.4	±0.8	±1.5	Gauge: 0.2–0.5
$T_{\max}$ (°C)	11.21	±0.6	±1.1	±2.2	Station: 0.2–0.5
$T_{\min}$ (°C)	10.27	±0.5	±1.0	±2.1	Station: 0.2–0.5
Radiation (W/m^2)	132.34	±6.6	±13.2	±26.5	Pyranometer: 5–10
Vap. pres. (Pa)	652.05	±32.6	±65.2	±130	Hygrometer: 20–50

2.5 Evaluation Metrics

Our primary robustness metric is $\Delta\text{NSE} = \text{NSE}_{\text{adv}} - \text{NSE}_{\text{clean}}$, where negative values indicate degradation caused by the perturbation. Because the per-basin distribution of ΔNSE is heavy-tailed toward harm, we report medians and interquartile ranges across catchments rather than means. Supplementary metrics include ΔKGE, peak-flow error, and a Kolmogorov–Smirnov (KS) two-sample p-value comparing the clean and perturbed marginal distributions of each forcing variable, used as a measure of how statistically detectable the perturbation would be under a routine distributional QC check.

All adversarial experiment blocks cover the same 531 benchmark basin IDs. The Nash–Sutcliffe Efficiency is, by construction, undefined when the observed-streamflow variance in an evaluation window is near zero, because NSE divides by that variance; 17 of the 531 basins fall into this degenerate regime in the test period and yield non-finite NSE and ΔNSE values. The raw streamflow and forcing data for these basins are complete; the metric simply cannot be computed stably on their evaluation windows. NSE-based headline statistics are therefore reported over the 514 basins with finite NSE and ΔNSE, while basin-coverage statements retain the full 531 wherever experiment provenance is documented.

2.6 Experimental Design

Experiments are organized into five blocks, each addressing a distinct robustness question across all 531 benchmark basins.

The *attack-comparison* block applies the five perturbation methods of Section 2.2 at four budget levels ($\varepsilon \in \{0.01, 0.05, 0.1, 0.2\}$) under the magnitude constraint and an untargeted loss that aggregates over all timesteps. This block isolates the effect of attack method choice. The *constraint-ablation* block holds the attack fixed at Auto-PGD and varies the constraint level (magnitude, physical, statistical) across three budgets ($\varepsilon \in \{0.05, 0.1, 0.2\}$), isolating the effect of plausibility constraints on achievable damage.

The *targeted-attack* block applies Auto-PGD at $\varepsilon = 0.1$ under three loss targets: an untargeted aggregate loss, a flood-targeted loss that concentrates the optimization on high-flow periods, and a low-flow-targeted loss that concentrates on dry periods. This block enables a direct comparison of high-flow and low-flow vulnerability. The *causal-trigger* block restricts perturbations to pre-flood windows of 1, 3, 7, and 14 days at $\varepsilon = 0.1$ and quantifies the share of vulnerability concentrated in the operational forecasting window.

The final *minimum-perturbation* block runs the C&W attack on each basin to estimate per-basin safety margins. All experiments were executed as chunked GPU jobs and written to an isolated 531-basin output tree. Results were merged and deduplicated, yielding 17,523 individual experiment records across 531 unique basin IDs.

3 Results

3.1 Attack-Method Comparison and Vulnerability Distribution

Table 3 reports median ΔNSE across the 514 finite basin metrics for each attack method at four perturbation budgets. Three features of these results merit comment: (i) the dominant gap is between methods that use the trained model's gradient (FGSM, Auto-PGD) and methods that perturb the forcing without reference to the model (the three random baselines), rather than between iterative and single-step gradient methods; (ii) the gap between single-step FGSM and iterative Auto-PGD increases monotonically with ε; and (iii) the underlying ΔNSE distribution is heavy-tailed, with the at-risk tail also widening with ε.

Table 3 — Median ΔNSE [Q25, Q75] across finite basin metrics for each attack method (magnitude constraint, untargeted). All rows cover 531 basin IDs; $N = 514$ denotes basins with finite NSE and ΔNSE.

Method	N	ε=0.01	ε=0.05	ε=0.1	ε=0.2
Auto-PGD	514	−0.026 [−0.044, −0.016]	−0.163 [−0.282, −0.092]	−0.408 [−0.731, −0.225]	−1.162 [−2.298, −0.567
FGSM	514	−0.026 [−0.043, −0.016]	−0.154 [−0.267, −0.088]	−0.358 [−0.640, −0.194]	−0.824 [−1.672, −0.436
Gaussian	514	−0.002 [−0.003, −0.001]	−0.010 [−0.018, −0.006]	−0.023 [−0.039, −0.012]	−0.054 [−0.095, −0.028
Mult. Bias	514	−0.001 [−0.003, −0.001]	−0.007 [−0.016, −0.003]	−0.015 [−0.035, −0.007]	−0.033 [−0.080, −0.016
Temp. Corr.	514	−0.002 [−0.003, −0.001]	−0.008 [−0.015, −0.004]	−0.016 [−0.031, −0.009]	−0.037 [−0.067, −0.020
APGD/Gauss		13.0×	15.6×	17.5×	21.6×

At $\varepsilon = 0.1$, the median ΔNSE under Auto-PGD is −0.408, compared to −0.358 for FGSM and −0.023 for Gaussian noise. All three random baselines cluster within a narrow band around −0.02 regardless of their temporal structure (Gaussian −0.023, multiplicative bias −0.015, temporally correlated noise −0.016), confirming that the dominant factor distinguishing methods is the use of gradient information rather than the temporal correlation of the perturbation. The Auto-PGD-to-Gaussian ratio at $\varepsilon = 0.1$ is 17.5×, more than an order of magnitude larger than the Auto-PGD-to-FGSM ratio of 1.14×.

The Auto-PGD-to-FGSM ratio grows monotonically with ε: 1.00× at $\varepsilon = 0.01$, 1.06× at $\varepsilon = 0.05$, 1.14× at $\varepsilon = 0.1$, and 1.41× at $\varepsilon = 0.2$ (Figure 1). At small budgets the model's response to forcing perturbations is approximately linear within the ε neighborhood, so a single gradient step nearly saturates the achievable damage; at larger budgets the model's response curves away from this linear approximation, and iterative refinement uncovers worse forcing-error patterns that the single-step probe cannot reach.

The Auto-PGD-to-Gaussian ratio also grows with ε (13.0×, 15.6×, 17.5×, 21.6×), so random-noise sensitivity analysis underestimates worst-case vulnerability most severely at the perturbation budgets that matter most for stress testing.

3.1.1 Distribution-Level View at $\varepsilon = 0.1$

Figure 2 shows the per-basin distribution of ΔNSE under Auto-PGD at $\varepsilon = 0.1$. The distribution is strongly right-skewed: the 10th percentile is −1.51, the median is −0.408, and the 90th percentile is −0.12. The distance between the 10th percentile and the median (1.10) is roughly 3.5 times the distance between the median and the 90th percentile (0.29), demonstrating that the median statistic alone hides a long tail of severely vulnerable basins.

The size of this tail differs sharply across attack types. The fraction of basins with adversarial NSE below zero is 32.5% under Auto-PGD, 28.8% under FGSM, and 7.4% under Gaussian noise. Under the stricter $\Delta\text{NSE} < -1$ threshold, Auto-PGD produces severe degradation in 90 of 514 finite basins (17.5%). Even the simplest gradient-based method (FGSM) identifies an at-risk population nearly four times larger than random noise, and iterative refinement adds a further 3.7 percentage points of catchments below NSE = 0. Random-noise testing therefore underestimates not only the typical degradation but also the size of the at-risk tail population.

3.2 Statistically Undetectable Perturbations Remain Effective

Table 4 presents the constraint-ablation results for Auto-PGD under the three plausibility levels. The physical constraint (clipping precipitation to non-negative values and bounding temperatures to climatologically reasonable ranges) reduces median degradation by only 9% relative to the magnitude-only baseline at $\varepsilon = 0.1$ (from −0.408 to −0.370), indicating that the most damaging adversarial forcing patterns are often already physically plausible without explicit enforcement: forcing the model into worst-case behavior does not require negative precipitation or unrealistic temperatures.

The statistical constraint, which additionally requires the perturbed forcing to preserve the per-variable mean and standard deviation of the clean series, reduces median degradation by 53% (to −0.193 at $\varepsilon = 0.1$). The larger reduction is intuitive: pinning the moments removes precisely the systematic shifts that adversarial perturbations would otherwise use to bias the LSTM's recurrent state.

Table 4 — Constraint ablation results for Auto-PGD (untargeted). Values are median (mean) ΔNSE over the 514 finite basin metrics.

Constraint	N	ε=0.05	ε=0.1	ε=0.2
Magnitude	514	−0.163 (−0.278)	−0.408 (−0.713)	−1.162 (−2.169)
Physical	514	−0.148 (−0.226)	−0.370 (−0.622)	−1.083 (−1.991)
Statistical	514	−0.083 (−0.113)	−0.193 (−0.258)	−0.459 (−0.633)

Crucially, the residual degradation under the statistical constraint is operationally significant and grows with ε: from −0.083 at $\varepsilon = 0.05$ to −0.193 at $\varepsilon = 0.1$ to −0.459 at $\varepsilon = 0.2$. The ratio of statistical-constraint to magnitude-only degradation stays near 0.4–0.5 across budgets, so moment-preserving perturbations recover roughly half of the unconstrained adversarial damage at every scale. The practical implication is direct: forcing data that passes routine QC by reproducing the climatological mean and standard deviation of each variable can still contain structured errors that substantially degrade model performance, and moment-based QC alone cannot rule out the existence of harmful forcing-error patterns within an observation-error-scale budget.

3.3 Targeted Attacks and Pre-Event Vulnerability

Table 5 presents the results of targeted attacks. Flood-targeted attacks produce degradation comparable to, though slightly less than, untargeted attacks, while low-flow-targeted attacks are much weaker. The flood/low-flow contrast is approximately 16:1 by median ΔNSE, indicating that vulnerability is concentrated in high-flow regimes.

Table 5 — Targeted attack results (Auto-PGD, $\varepsilon = 0.1$, magnitude constraint). Median [Q25, Q75] across finite basin metrics. ΔNSE uses $N = 514$; ΔKGE uses all 531 basin IDs.

Target	N	Median ΔNSE [IQR]	Median ΔKGE [IQR]
Untargeted	514	−0.408 [−0.731, −0.225]	−0.380 [−0.819, −0.133]
Flood	514	−0.332 [−0.602, −0.182]	−0.344 [−0.716, −0.132]
Low-flow	514	−0.021 [−0.092, 0.003]	−0.049 [−0.236, 0.024]

Figure 3 shows the causal-trigger results. Restricting perturbations to progressively longer pre-event windows yields a monotonic increase in degradation: 1 day (−0.011), 3 days (−0.028), 7 days (−0.050), and 14 days (−0.085). Even perturbations confined to the 14 days preceding flood peaks cause non-trivial degradation concentrated in the operationally critical forecasting window.

3.4 Safety Margins and Detectability

The previous subsections quantified how strongly the model can be degraded under fixed perturbation budgets. Two complementary aspects characterize how silently this damage can occur: the minimum perturbation magnitude required to break the model, and the detectability of harmful perturbations under standard distribution tests on the forcings. A small required perturbation that is also statistically invisible defines the worst case for input-side defenses.

The Carlini–Wagner (C&W) attack identifies, for each basin, the minimum L_2 perturbation required to drive NSE below zero (Figure 4). Of 531 basin IDs evaluated, 369 (69.5%) yielded non-negligible solutions ($L_2 > 0.01$) with a median minimum perturbation of 1.244 in standardized units. The remaining 162 basins returned near-zero values, reflecting either an already low safety margin (the clean prediction is close to NSE = 0 to begin with) or failed convergence in the minimum-perturbation search rather than actual immunity. The median safety margin is on the same order as the standardized $\varepsilon = 1$ ball, but a non-trivial fraction of basins can be broken with $L_2 \ll 1$, particularly those with weaker baseline performance.

Detectability is assessed by computing a Kolmogorov–Smirnov (KS) two-sample test between the clean and perturbed input distributions for each forcing variable, and pairing the resulting p-value with the achieved |ΔNSE|. Figure 5 shows the joint distribution. Adversarial methods achieve large degradation across a wide range of p-values, including many cases with $p > 0.05$ that would not be flagged as statistically anomalous by a standard two-sample test. Effective adversarial perturbations therefore do not require statistical anomaly: they exploit the temporal-gradient structure of the model rather than violating the marginal input distributions, which is consistent with the constraint-ablation result that moment-preserving perturbations retain roughly half of the unconstrained damage.

Combining the two analyses, harmful perturbations exist whose L_2 magnitude is on the order of one standardized unit and whose marginal distributions remain statistically indistinguishable from the clean forcings. This is the worst case for any QC strategy that monitors only the forcing data, because both the magnitude and the marginal-distribution signatures of these perturbations fall within the range that routine input checks would consider acceptable. Detecting them would require diagnostics that incorporate output from the trained model itself, such as residual checks on the predicted streamflow against an independent reference, or training-time defenses such as adversarial training, rather than forcing-side distribution tests alone.

4 Discussion

4.1 Relation to Prior Adversarial Assessment of Hydrological LSTMs

The closest precedent for this study is the FGSM-based robustness analysis of Yang et al. (2026), who applied single-step adversarial perturbations to LSTM and HBV rainfall-runoff models across 1,347 CAMELS-DE catchments. Their headline findings were that LSTMs were more robust than process-based HBV models, that catastrophic failure was rare, and that model responses scaled approximately linearly with perturbation magnitude. The 531-basin CAMELS-US results presented here qualify rather than contradict that picture.

At the same observation-error-scale budget ($\varepsilon = 0.1$), FGSM on our victim model produces a median ΔNSE of -0.358 and pushes 28.8% of finite basins below NSE $= 0$. Iterative Auto-PGD moves the median to -0.408 and the at-risk fraction to 32.5%, with a further 17.5% of basins falling below ΔNSE $= -1$. Two of the three qualitative claims of Yang et al. (2026) survive: random noise remains a much weaker probe than gradient-based attack, and the response to FGSM is approximately monotonic in ε across our budget range. The third claim—that catastrophic failure is rare—is harder to defend once iterative methods and larger budgets are admitted. Severe degradation under Auto-PGD at $\varepsilon = 0.1$ affects roughly one in three finite basins; it is a population-level feature rather than a tail of curiosities.

The methodological complement is consistent with the broader adversarial machine-learning literature, in which single-step methods are documented to systematically overestimate model robustness (Carlini et al., 2019). The FGSM-to-Auto-PGD ratio observed here (1.14$\times$ at $\varepsilon = 0.1$, growing to 1.41$\times$ at $\varepsilon = 0.2$) transposes that pattern from image classification to a regression task on physical time series.

4.2 Mechanisms Underlying the Attack-Method Hierarchy

The hierarchy among the five methods at $\varepsilon = 0.1$—Auto-PGD followed by FGSM, with the three random baselines clustered near zero—admits a unified explanation. Random methods perturb each timestep and forcing variable independently and are exponentially unlikely to align with the directions of greatest model sensitivity. FGSM uses the gradient at the clean operating point to identify an aligned direction but takes only one step, so in regions where the model's response curves within the ε neighborhood, this direction is no longer the worst attainable perturbation. Auto-PGD iteratively tracks the curvature, exploiting nonlinearities that FGSM cannot reach.

The adversarial-to-random gap (17.5$\times$ at $\varepsilon = 0.1$) far exceeds the FGSM-to-Auto-PGD gap (1.14$\times$), revealing the relative importance of the two methodological refinements: using gradient information at all matters far more than refining how that gradient is followed. The corresponding methodological recommendation is asymmetric: a single-step gradient-based attack should be a minimum requirement for any future hydrological robustness assessment, whereas iterative refinement is the natural gold standard for deployment-grade evaluations.

The constraint-ablation experiment provides a further mechanistic clue. Statistical constraints—which require the perturbed forcing to preserve per-variable mean and standard deviation—halve the median damage relative to the unconstrained magnitude baseline, but more than half of the damage remains. The residual damage cannot be carried by changes in the marginal moments of the inputs, since those moments are pinned by construction; it must instead be carried by changes in *temporal sequencing* (the order in which dry and wet days follow one another) and *cross-variable correlations* (the joint pattern of precipitation, temperature, and radiation) to which the LSTM's recurrent state is sensitive. Adversarial perturbations exploit this temporal and cross-variable structure rather than the absolute scale of the inputs, which explains why moment-based detectability checks (Section 3.4) cannot reliably flag them.

4.3 Implications for Operational Hydrological Deployment

These findings should not be read as evidence against the operational deployment of LSTM rainfall-runoff models. The adversarial perturbations are diagnostic rather than predictive: they test whether harmful error structures exist within observation-error-scale uncertainty budgets, not whether those exact perturbations are likely to occur in practice. A model that fails under such constrained stress tests cannot, however, be assumed robust on the basis of average-case random-noise experiments alone, because the stress test reveals harm that the random analysis is structurally unable to find.

A first implication concerns reporting conventions. Hydrological deep-learning studies routinely report clean-data NSE, KGE, and percentile error statistics across catchments. We suggest that an adversarial ΔNSE at $\varepsilon = 0.1$ should accompany these summaries whenever a model is proposed for operational use. The marginal cost is modest—a single Auto-PGD pass per catchment over the test period—and the diagnostic power, as the Results above demonstrate, is qualitatively beyond what random-noise sensitivity analysis can provide.

A second implication concerns input quality control. Standard QC pipelines built around marginal-moment checks are well-suited to detecting gross sensor failure but are blind by construction to the moment-preserving perturbations isolated in the constraint-ablation experiment, even though such perturbations retain roughly half of the unconstrained adversarial damage at every budget tested. Detecting them would require higher-order input validation—temporal autocorrelation checks, cross-feature consistency tests, or model-side reconstruction residuals—rather than purely distributional checks at the feature-marginal level.

A third implication concerns the temporal structure of forecasting risk. The causal-trigger experiment shows that perturbations restricted to the fourteen days immediately preceding a flood peak still produce a median ΔNSE of -0.085. In operational flood forecasting this is precisely the window in which input data are most consequential and most often degraded by sensor failure or telemetry latency. Robustness budgets and data-quality monitoring should therefore be tightened around such pre-event windows rather than applied uniformly across the entire time series.

4.4 Caveats and Avenues for Future Work

Several caveats bound the generality of these conclusions. The victim model is a single-layer LSTM with 128 hidden units and a fixed set of static catchment attributes. Entity-aware LSTMs, sequence Transformers, and state-space architectures such as Mamba may interact with adversarial perturbations differently; a systematic architectural sweep on the same 531-basin benchmark is the natural next step. The dataset is CAMELS-US at daily resolution. Sub-daily datasets and non-CONUS hydroclimates, particularly those with stronger seasonal precipitation regimes, may shift both the median and the tail of the vulnerability distribution. Within the 531-basin benchmark itself, 17 basins yielded non-finite NSE-based robustness metrics because their evaluation windows lacked sufficient observed-flow variability for stable NSE calculation. Headline statistics are therefore reported over the 514 finite basins, while basin-coverage statements retain the full 531.

Defense strategies were intentionally outside the scope of this study. Adversarial training, ensemble averaging across model seeds, and input-side denoising each make a different methodological commitment, and a rigorous evaluation of these strategies—ideally against the same multi-level stress test rather than against FGSM alone—is a natural follow-up. The fact that statistically constrained perturbations retain about half of the unconstrained damage suggests that input-side denoising is unlikely to suffice on its own.

A final interpretive caveat is the diagnostic-not-predictive framing noted above. The adversarial perturbations here provide a worst-case bound within an observation-error-scale budget; they do not estimate the empirical frequency of such forcing-error patterns in operational settings. Closing that gap would require either coupling the attack to a calibrated input-error model or comparing the adversarial ΔNSE distribution to the ΔNSE distribution induced by historical input-error events. Both are open directions for future work.

5 Conclusions

We investigated how the choice of perturbation method affects adversarial robustness conclusions for a standard LSTM rainfall-runoff configuration, evaluating five methods spanning the spectrum from random noise to iterative gradient-based optimization across the 531-basin CAMELS-US temporal benchmark.

The 531-basin CAMELS-US benchmark experiments support four principal findings. First, the choice of attack method exerts a decisive influence on the robustness conclusions drawn for a standard LSTM rainfall-runoff configuration: iterative adversarial perturbations produce a median degradation 17.5 times greater than Gaussian random noise at $\varepsilon = 0.1$, and FGSM, although closer to Auto-PGD than to the random baselines, still systematically underestimates the worst case at larger budgets. Second, vulnerability is not a tail phenomenon: under Auto-PGD at $\varepsilon = 0.1$, 32.5% of finite basins fall below NSE $= 0$ and 17.5% exhibit severe degradation with ΔNSE < -1. Third, perturbations that preserve the per-variable mean and standard deviation of the forcing retain roughly half of the unconstrained adversarial damage at every budget tested, indicating that routine moment-based quality control is insufficient to rule out harmful forcing patterns. Fourth, pre-event data quality is disproportionately important: perturbations confined to the fourteen days preceding a flood peak still produce a median ΔNSE of -0.085, in the operational forecasting window in which input data are most consequential and most often degraded.

Taken together, these findings indicate that robustness assessment for standard hydrological deep-learning configurations requires multi-level stress testing that extends beyond random noise and single-step adversarial probes. We recommend iterative adversarial evaluation at observation-error-scale budgets as a standard diagnostic component of hydrological stress-testing workflows. Broader architectural generalization, evaluation of defense strategies, and the coupling of adversarial bounds to calibrated forcing-error models remain natural follow-ups.

Open Research

All code for adversarial attacks, evaluation, and analysis is available at `https://github.com/[repository]`. The LSTM model was trained using the open-source neuralhydrology framework (Kratzert et al., 2022). CAMELS-US data are publicly available from the USGS (Newman et al., 2015; Addor et al., 2017).

Acknowledgments

[Acknowledgments to be added.]

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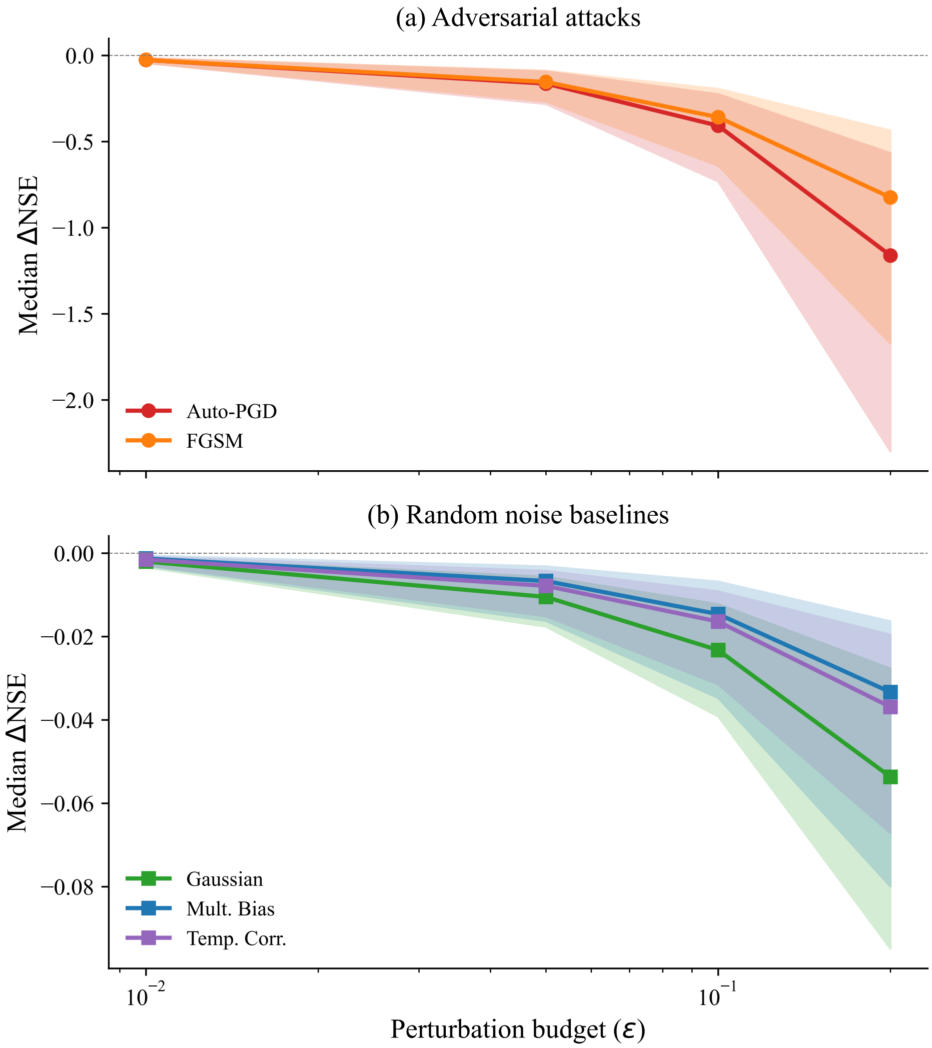

Figure 1 — Median ΔNSE as a function of ε for five perturbation methods, with IQR shown as shaded bands. (a) Adversarial attacks: Auto-PGD and FGSM. (b) Random noise baselines. Note the different y-axis scales between panels.

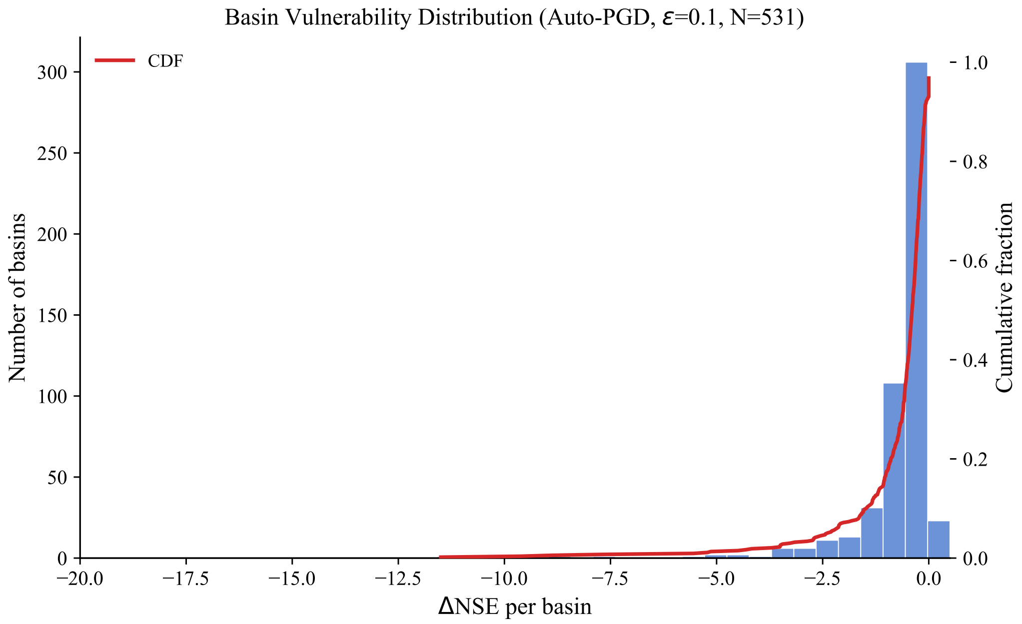

Figure 2 — Distribution of per-catchment ΔNSE under Auto-PGD at $\varepsilon = 0.1$ ($N = 514$ finite NSE metrics from 531 basin IDs). Dashed lines mark the 10th percentile (−1.51), median (−0.41), and 90th percentile (−0.12).

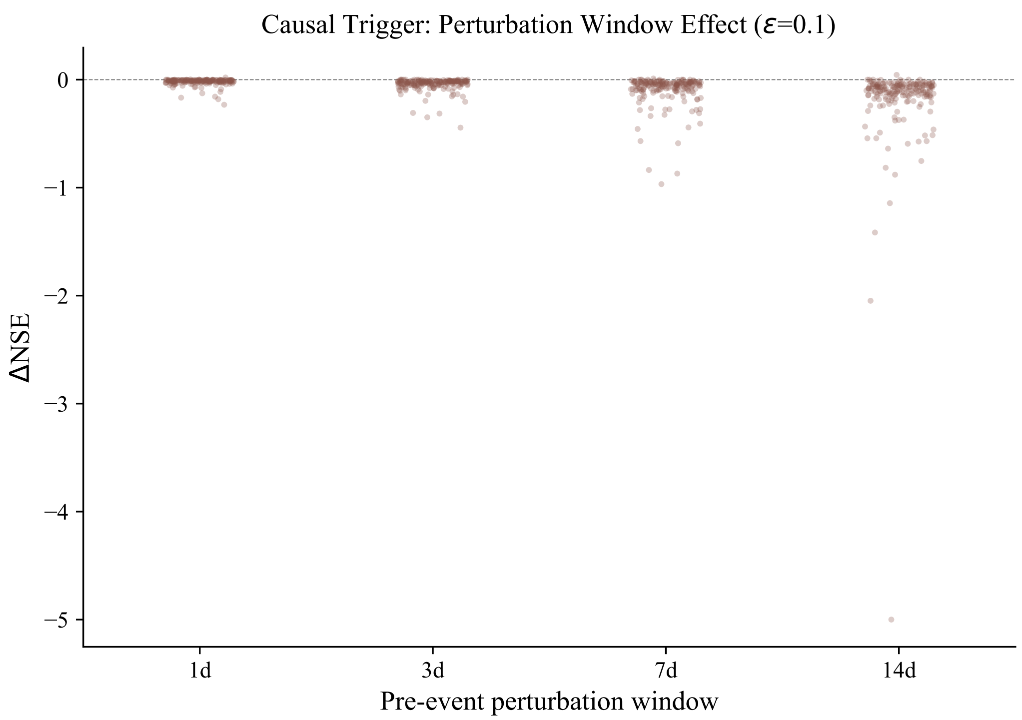

Figure 3 — Causal trigger: pre-event window length versus ΔNSE ($\varepsilon = 0.1$). Box plots with jittered dots. Median values are annotated.

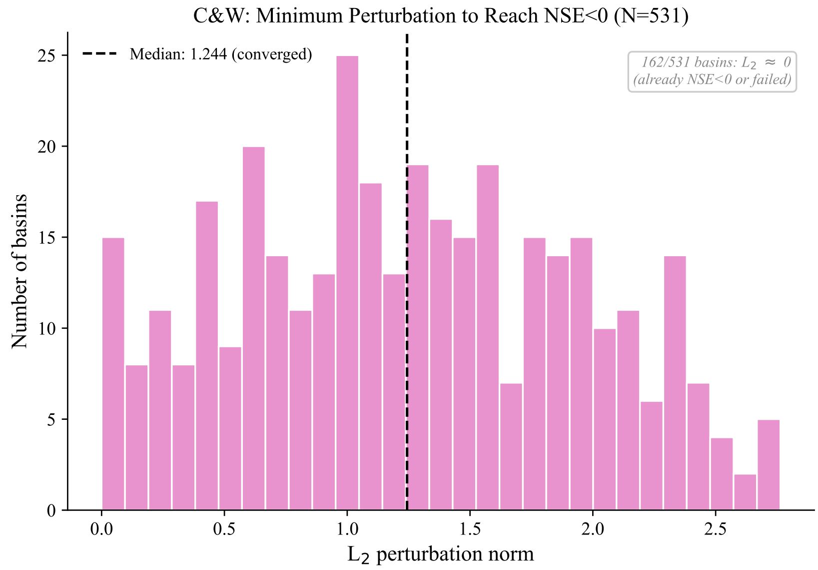

Figure 4 — Distribution of minimum L_2 perturbation required to reduce NSE below zero ($N = 531$ basin IDs). Non-negligible solutions use the analysis threshold $L_2 > 0.01$ ($N = 369$); 162 near-zero cases are annotated.

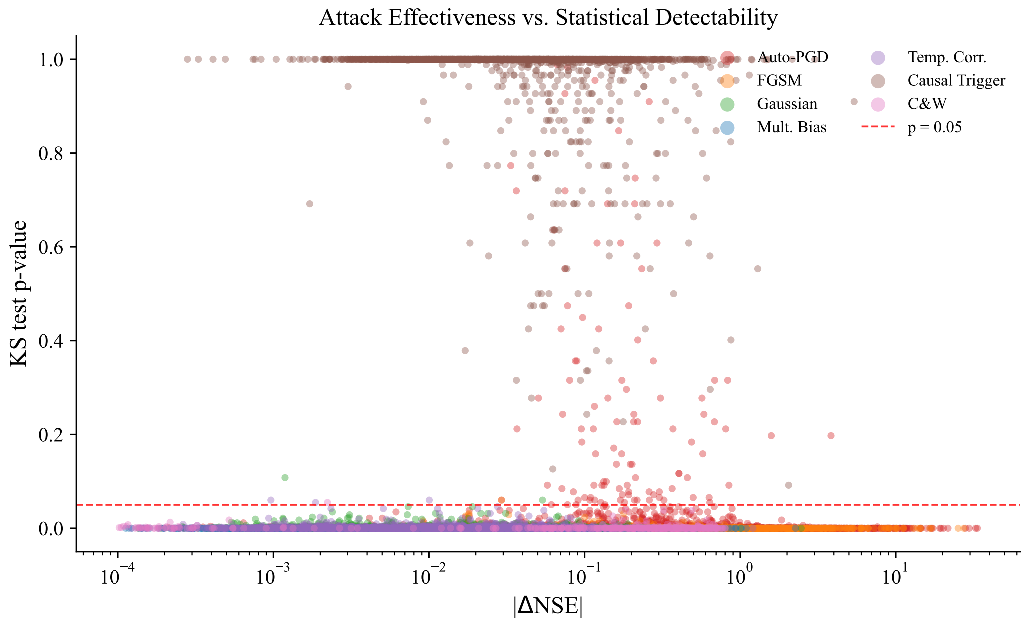

Figure 5 — Attack effectiveness (|ΔNSE|) versus statistical detectability (KS p-value). The red dashed line marks $p = 0.05$.